

Pràctica Símplex: Resolució de Problemes

1. Membres del grup

- Carlos Palazón - DNI:
- Pol Riera - DNI: 39480980P

2. Nombre de conjunts de dades usats

S'han utilitzat **dos conjunts de dades**: el conjunt **34** i el conjunt **37**, resolent un total de 8 problemes d'optimització.

3. Descripció resumida de la implementació

Hem implementat la solució en Python a través de la classe `SimplexSolver`, aplicant el **mètode del símplex primal estructurat en dues fases** i utilitzant la **regla de Bland** en totes les decisions per evitar la degeneració.

La seqüència d'execució del nostre codi és la següent:

1. **Preprocessament**: Forcem els termes independents a ser no negatius ($b_i \geq 0$) canviant el signe de tota la fila corresponent de la matriu A si és necessari.
2. **Fase I (Solució Bàsica Factible Inicial)**: Per assegurar l'arrencada independentment de les restriccions del problema, estenem la matriu afegint variables artificials (una matriu identitat) que componen la base inicial. Executem el nucli del símplex amb una funció objectiu que minimitza la suma d'aquestes variables artificials. Si el valor òptim d'aquesta fase és major que 0 (considerant una tolerància de 10^{-8}), declarem el problema com a infactible.
3. **Fase II (Optimització real)**: Un cop eliminades les variables artificials no bàsiques i garantida la factibilitat, restaurem el vector de costos original c i reprenem l'execució del símplex fins assolir l'òptim global o detectar no acotació.
4. **Nucli iteratiu**: En cada iteració, per calcular valors de variables bàsiques (x_B), multiplicadors i la direcció de descens, resollem els sistemes d'equacions lineals associats $Bx_B = b$, $B^T w = c_B$ i

$Bd = A_q$ aplicant el mètode de **Gauss-Jordan amb pivoteig parcial** per garantir estabilitat numèrica.

5. **Pivoteig (Regla de Bland):** Per entrar a la base calculem els costos reduïts (r_j) manualment; d'entre els que són negatius (més enllà del llindar de tolerància), seleccionem el candidat amb l'índex j més petit. Per sortir de la base apliquem el test del quocient $\min\{x_B/d\}$ considerant $d > 0$, escollint novament l'índex més baix en cas d'empat per complir amb la Regla de Bland.

4. Solució obtinguda

A continuació es detalla el registre de l'execució pas a pas de cada iteració i les solucions òptimes per als 8 problemes dels conjunts indicats.

Problema 1

[jh_simplexP] Fase I

```
[jh_simplexP] Solucio basica factible trobada, iteracio 19
```

[jh_simplexP]	Iteracio	20 : iout =	6, q =	5, B(p) =	9, theta*=	1.706, z =	-77.097
[jh_simplexP]	Iteracio	21 : iout =	5, q =	15, B(p) =	8, theta*=	179.166, z =	-132.516
[jh_simplexP]	Iteracio	22 : iout =	9, q =	16, B(p) =	4, theta*=	9.381, z =	-245.214
[jh_simplexP]	Iteracio	23 : iout =	5, q =	1, B(p) =	15, theta*=	1.397, z =	-251.992
[jh_simplexP]	Iteracio	24 : iout =	6, q =	8, B(p) =	5, theta*=	0.702, z =	-296.160
[jh_simplexP]	Iteracio	25 : iout =	7, q =	17, B(p) =	2, theta*=	273.616, z =	-323.146
[jh_simplexP]	Iteracio	26 : iout =	1, q =	19, B(p) =	6, theta*=	147.251, z =	-363.306
[jh_simplexP]	Iteracio	27 : iout =	5, q =	2, B(p) =	1, theta*=	0.761, z =	-531.478
[jh_simplexP]	Iteracio	28 : iout =	7, q =	5, B(p) =	17, theta*=	0.832, z =	-580.361
[jh_simplexP]	Iteracio	29 : iout =	3, q =	7, B(p) =	11, theta*=	1.360, z =	-588.083
[jh_simplexP]	Iteracio	30 : iout =	7, q =	17, B(p) =	5, theta*=	356.161, z =	-630.546
[jh_simplexP]	Iteracio	31 : iout =	5, q =	20, B(p) =	2, theta*=	104.747, z =	-639.085
[jh_simplexP]	Iteracio	32 : iout =	2, q =	5, B(p) =	13, theta*=	2.203, z =	-695.736
[jh_simplexP]	Iteracio	33 : iout =	7, q =	1, B(p) =	17, theta*=	0.004, z =	-787.471
[jh_simplexP]	Iteracio	34 : iout =	-, q =	-, B(p) =	-, theta*=	-, z =	-787.487

```
[jh simplexP]      Solucio optima trobada, iteracio 34, z = -787.486630
```

[jh_simplexP] Fi simplex primal

Solucio optima:

vb = 19 5 7 10 20 8 1 12 16 3

xb = 764.4 2.2 1.5 2.5 291.8 3.6 0.0 2.2 322.8 0.5

z = -787.4866

r = 0.0 31.4 0.0 110.0 -0.0 192.2 -0.0 0.0 176.1 -0.0 5.5 0.0 114.3

Problema 2

[jh_simplexP] Inici simplex primal amb regla de Bland

[jh_simplexP] Fase I

[jh_simplexP]	Iteracio	1	:	iout =	1,	q =	1,	B(p) =	21,	theta*=	0.113,	z =	2231.000
[jh_simplexP]	Iteracio	2	:	iout =	8,	q =	3,	B(p) =	28,	theta*=	0.030,	z =	2223.189
[jh_simplexP]	Iteracio	3	:	iout =	6,	q =	4,	B(p) =	26,	theta*=	0.951,	z =	2208.520
[jh_simplexP]	Iteracio	4	:	iout =	10,	q =	2,	B(p) =	30,	theta*=	0.145,	z =	1766.561
[jh_simplexP]	Iteracio	5	:	iout =	5,	q =	5,	B(p) =	25,	theta*=	2.494,	z =	1724.330
[jh_simplexP]	Iteracio	6	:	iout =	1,	q =	6,	B(p) =	1,	theta*=	0.132,	z =	1441.831
[jh_simplexP]	Iteracio	7	:	iout =	1,	q =	7,	B(p) =	6,	theta*=	0.108,	z =	1412.946
[jh_simplexP]	Iteracio	8	:	iout =	9,	q =	8,	B(p) =	29,	theta*=	0.008,	z =	1393.373
[jh_simplexP]	Iteracio	9	:	iout =	1,	q =	6,	B(p) =	7,	theta*=	0.666,	z =	1379.835
[jh_simplexP]	Iteracio	10	:	iout =	4,	q =	10,	B(p) =	24,	theta*=	0.394,	z =	1314.303
[jh_simplexP]	Iteracio	11	:	iout =	8,	q =	1,	B(p) =	3,	theta*=	2.818,	z =	894.204
[jh_simplexP]	Iteracio	12	:	iout =	10,	q =	7,	B(p) =	2,	theta*=	1.790,	z =	888.772
[jh_simplexP]	Iteracio	13	:	iout =	8,	q =	3,	B(p) =	1,	theta*=	1.718,	z =	411.338
[jh_simplexP]	Iteracio	14	:	iout =	2,	q =	9,	B(p) =	22,	theta*=	0.237,	z =	392.790
[jh_simplexP]	Iteracio	15	:	iout =	3,	q =	11,	B(p) =	23,	theta*=	0.050,	z =	313.545
[jh_simplexP]	Iteracio	16	:	iout =	2,	q =	1,	B(p) =	9,	theta*=	0.049,	z =	282.348
[jh_simplexP]	Iteracio	17	:	iout =	6,	q =	2,	B(p) =	4,	theta*=	1.079,	z =	278.716
[jh_simplexP]	Iteracio	18	:	iout =	7,	q =	13,	B(p) =	27,	theta*=	1.030,	z =	136.476
[jh_simplexP]	Iteracio	19	:	iout =	-,	q =	-,	B(p) =	-,	theta*=	-,	z =	0.000

[jh_simplexP] Solucio basica factible trobada, iteracio 19

[jh_simplexP] Fase II

[jh_simplexP]	Iteracio	20	:	iout =	10,	q =	9,	B(p) =	7,	theta*=	2.023,	z =	417.139
[jh_simplexP]	Iteracio	21	:	iout =	1,	q =	12,	B(p) =	6,	theta*=	2.362,	z =	267.479
[jh_simplexP]	Iteracio	22	:	iout =	8,	q =	7,	B(p) =	3,	theta*=	0.057,	z =	135.921
[jh_simplexP]	Iteracio	23	:	iout =	5,	q =	14,	B(p) =	5,	theta*=	0.100,	z =	119.775
[jh_simplexP]	Iteracio	24	:	iout =	8,	q =	3,	B(p) =	7,	theta*=	0.520,	z =	108.443
[jh_simplexP]	Iteracio	25	:	iout =	9,	q =	4,	B(p) =	8,	theta*=	0.316,	z =	82.939
[jh_simplexP]	Iteracio	26	:	iout =	9,	q =	6,	B(p) =	4,	theta*=	0.180,	z =	56.912
[jh_simplexP]	Iteracio	27	:	iout =	9,	q =	15,	B(p) =	6,	theta*=	105.586,	z =	54.896
[jh_simplexP]	Iteracio	28	:	iout =	4,	q =	16,	B(p) =	10,	theta*=	73.570,	z =	53.658
[jh_simplexP]	Iteracio	29	:	iout =	10,	q =	4,	B(p) =	9,	theta*=	1.514,	z =	18.470
[jh_simplexP]	Iteracio	30	:	iout =	10,	q =	6,	B(p) =	4,	theta*=	0.622,	z =	-23.129
[jh_simplexP]	Iteracio	31	:	iout =	9,	q =	10,	B(p) =	15,	theta*=	0.926,	z =	-32.778
[jh_simplexP]	Iteracio	32	:	iout =	9,	q =	17,	B(p) =	10,	theta*=	150.172,	z =	-42.298
[jh_simplexP]	Iteracio	33	:	iout =	8,	q =	19,	B(p) =	3,	theta*=	444.748,	z =	-93.051
[jh_simplexP]	Iteracio	34	:	iout =	-,	q =	-,	B(p) =	-,	theta*=	-,	z =	-250.134

[jh_simplexP] Solucio optima trobada, iteracio 34, z = -250.133658

[jh_simplexP] Fi simplex primal

Solucio optima:

```
vb = 12  1 11 16 14  2 13 19 17  6
xb =   1.4   2.4   0.3 294.0   4.3   0.1   3.0 444.7 520.0   2.7
z = -250.1337
r =   0.0  -0.0  90.3  16.0  32.7   0.0 189.2  91.8  48.4  57.6   0.0   0.0  -0.0
```

Problema 3

```
[jh_simplexP] Inici simplex primal amb regla de Bland
[jh_simplexP] Fase I
[jh_simplexP] Iteracio 1 : iout = 8, q = 1, B(p) = 28, theta*= 0.362, z = 1121.000
[jh_simplexP] Iteracio 2 : iout = 10, q = 2, B(p) = 30, theta*= 0.072, z = 1107.950
[jh_simplexP] Iteracio 3 : iout = 8, q = 3, B(p) = 1, theta*= 0.162, z = 1103.248
[jh_simplexP] Iteracio 4 : iout = 10, q = 4, B(p) = 2, theta*= 0.056, z = 1052.098
[jh_simplexP] Iteracio 5 : iout = 6, q = 5, B(p) = 26, theta*= 0.487, z = 1031.453
[jh_simplexP] Iteracio 6 : iout = 10, q = 1, B(p) = 4, theta*= 0.008, z = 951.743
[jh_simplexP] Iteracio 7 : iout = 10, q = 2, B(p) = 1, theta*= 0.006, z = 950.516
[jh_simplexP] Iteracio 8 : iout = 10, q = 8, B(p) = 2, theta*= 0.006, z = 950.032
[jh_simplexP] Iteracio 9 : iout = 10, q = 16, B(p) = 8, theta*= 0.809, z = 949.273
[jh_simplexP] Iteracio 10 : iout = 7, q = 17, B(p) = 27, theta*= 293.523, z = 948.026
[jh_simplexP] Iteracio 11 : iout = 10, q = 12, B(p) = 16, theta*= 0.112, z = 654.497
[jh_simplexP] Iteracio 12 : iout = 8, q = 18, B(p) = 3, theta*= 11.785, z = 638.813
[jh_simplexP] Iteracio 13 : iout = 9, q = 19, B(p) = 29, theta*= 48.817, z = 621.054
[jh_simplexP] Iteracio 14 : iout = 6, q = 26, B(p) = 5, theta*= 2.444, z = 572.237
[jh_simplexP] Iteracio 15 : iout = -, q = -, B(p) = -, theta*= -, z = 471.778
[jh_simplexP] Problema infactible.
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Problema 4

[jh_simplexP] Inici simplex primal amb regla de Bland

[jh_simplexP] Fase I

[jh_simplexP]	Iteracio	1	:	iout =	8,	q =	1,	B(p) =	32,	theta* =	8.155,	z =	7315.000
[jh_simplexP]	Iteracio	2	:	iout =	7,	q =	2,	B(p) =	31,	theta* =	2.340,	z =	3767.732
[jh_simplexP]	Iteracio	3	:	iout =	9,	q =	3,	B(p) =	33,	theta* =	6.480,	z =	2774.501
[jh_simplexP]	Iteracio	4	:	iout =	2,	q =	5,	B(p) =	26,	theta* =	0.965,	z =	2305.679
[jh_simplexP]	Iteracio	5	:	iout =	6,	q =	4,	B(p) =	30,	theta* =	0.222,	z =	1933.786
[jh_simplexP]	Iteracio	6	:	iout =	9,	q =	8,	B(p) =	3,	theta* =	0.099,	z =	392.172
[jh_simplexP]	Iteracio	7	:	iout =	6,	q =	6,	B(p) =	4,	theta* =	0.225,	z =	372.775
[jh_simplexP]	Iteracio	8	:	iout =	6,	q =	7,	B(p) =	6,	theta* =	1.002,	z =	368.020
[jh_simplexP]	Iteracio	9	:	iout =	3,	q =	9,	B(p) =	27,	theta* =	0.214,	z =	313.193
[jh_simplexP]	Iteracio	10	:	iout =	1,	q =	10,	B(p) =	25,	theta* =	0.312,	z =	276.343
[jh_simplexP]	Iteracio	11	:	iout =	5,	q =	3,	B(p) =	29,	theta* =	1.031,	z =	156.435
[jh_simplexP]	Iteracio	12	:	iout =	5,	q =	6,	B(p) =	3,	theta* =	0.486,	z =	110.614
[jh_simplexP]	Iteracio	13	:	iout =	10,	q =	11,	B(p) =	34,	theta* =	0.251,	z =	61.507
[jh_simplexP]	Iteracio	14	:	iout =	5,	q =	3,	B(p) =	6,	theta* =	0.593,	z =	27.792
[jh_simplexP]	Iteracio	15	:	iout =	4,	q =	12,	B(p) =	28,	theta* =	0.423,	z =	25.700
[jh_simplexP]	Iteracio	16	:	iout =	-	q =	-	B(p) =	-	theta* =	-	z =	0.000

[jh_simplexP] Solucio basica factible trobada, iteracio 16

[jh_simplexP] Fase II

[jh_simplexP]	Iteracio	17	:	iout =	5,	q =	6,	B(p) =	3,	theta* =	0.687,	z =	-774.572
[jh_simplexP]	Iteracio	18	:	iout =	3,	q =	13,	B(p) =	9,	theta* =	0.924,	z =	-894.881
[jh_simplexP]	Iteracio	19	:	iout =	1,	q =	14,	B(p) =	10,	theta* =	0.423,	z =	-982.164
[jh_simplexP]	Iteracio	20	:	iout =	4,	q =	15,	B(p) =	12,	theta* =	2.057,	z =	-1072.64
[jh_simplexP]	Iteracio	21	:	iout =	4,	q =	16,	B(p) =	15,	theta* =	1.275,	z =	-1073.65
[jh_simplexP]	Iteracio	22	:	iout =	6,	q =	18,	B(p) =	7,	theta* =	18.095,	z =	-1074.43
[jh_simplexP]	Iteracio	23	:	iout =	5,	q =	9,	B(p) =	6,	theta* =	0.307,	z =	-1088.04
[jh_simplexP]	Iteracio	24	:	iout =	7,	q =	15,	B(p) =	2,	theta* =	45.483,	z =	-1103.17
[jh_simplexP]	Iteracio	25	:	iout =	5,	q =	6,	B(p) =	9,	theta* =	0.122,	z =	-1122.54
[jh_simplexP]	Iteracio	26	:	iout =	5,	q =	20,	B(p) =	6,	theta* =	5.499,	z =	-1123.20
[jh_simplexP]	Iteracio	27	:	iout =	7,	q =	2,	B(p) =	15,	theta* =	1.014,	z =	-1128.11
[jh_simplexP]	Iteracio	28	:	iout =	9,	q =	17,	B(p) =	8,	theta* =	91.735,	z =	-1132.58
[jh_simplexP]	Iteracio	29	:	iout =	7,	q =	15,	B(p) =	2,	theta* =	200.097,	z =	-1183.09
[jh_simplexP]	Iteracio	30	:	iout =	10,	q =	7,	B(p) =	11,	theta* =	3.975,	z =	-1229.46
[jh_simplexP]	Iteracio	31	:	iout =	8,	q =	19,	B(p) =	1,	theta* =	198.423,	z =	-1259.7
[jh_simplexP]	Iteracio	32	:	iout =	10,	q =	11,	B(p) =	7,	theta* =	0.361,	z =	-1335.32
[jh_simplexP]	Iteracio	33	:	iout =	10,	q =	21,	B(p) =	11,	theta* =	30.122,	z =	-1337.27
[jh_simplexP]	Iteracio	34	:	iout =	3,	q =	22,	B(p) =	13,	theta* =	1934.565,	z =	-1383.6
[jh_simplexP]	Iteracio	35	:	iout =	2,	q =	7,	B(p) =	5,	theta* =	6.267,	z =	-2468.02
[jh_simplexP]	Iteracio	36	:	iout =	2,	q =	24,	B(p) =	7,	theta* =	564.000,	z =	-2594.6
[jh_simplexP]	Iteracio	37	:	iout =	3,	q =	13,	B(p) =	22,	theta* =	85.241,	z =	-4625.00

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[jh_simplexP] Iteracio 38 : iout = 1, q = 23, B(p) = 14, theta*= 631.500, z = -5477.41
[jh_simplexP] Iteracio 39 : problema no acotat detectat.
[jh_simplexP] Problema no acotat.
```

Conjunt 37

Problema 1

[jh_simplexP] Inici simplex primal amb regla de Bland

[jh_simplexP] Fase I

[jh_simplexP]	Iteracio	1	:	iout =	7,	q =	1,	B(p) =	27,	theta*=	0.072,	z =	2278.000
[jh_simplexP]	Iteracio	2	:	iout =	6,	q =	2,	B(p) =	26,	theta*=	3.391,	z =	2261.402
[jh_simplexP]	Iteracio	3	:	iout =	3,	q =	4,	B(p) =	23,	theta*=	1.038,	z =	1709.298
[jh_simplexP]	Iteracio	4	:	iout =	2,	q =	3,	B(p) =	22,	theta*=	0.038,	z =	1382.148
[jh_simplexP]	Iteracio	5	:	iout =	2,	q =	5,	B(p) =	3,	theta*=	0.035,	z =	1366.560
[jh_simplexP]	Iteracio	6	:	iout =	10,	q =	6,	B(p) =	30,	theta*=	0.166,	z =	1361.862
[jh_simplexP]	Iteracio	7	:	iout =	7,	q =	3,	B(p) =	1,	theta*=	0.183,	z =	1325.968
[jh_simplexP]	Iteracio	8	:	iout =	1,	q =	7,	B(p) =	21,	theta*=	0.663,	z =	1291.896
[jh_simplexP]	Iteracio	9	:	iout =	10,	q =	1,	B(p) =	6,	theta*=	0.012,	z =	1106.070
[jh_simplexP]	Iteracio	10	:	iout =	9,	q =	9,	B(p) =	29,	theta*=	1.593,	z =	1099.446
[jh_simplexP]	Iteracio	11	:	iout =	5,	q =	10,	B(p) =	25,	theta*=	0.773,	z =	584.210
[jh_simplexP]	Iteracio	12	:	iout =	8,	q =	11,	B(p) =	28,	theta*=	1.221,	z =	277.690
[jh_simplexP]	Iteracio	13	:	iout =	10,	q =	6,	B(p) =	1,	theta*=	0.127,	z =	65.586
[jh_simplexP]	Iteracio	14	:	iout =	10,	q =	8,	B(p) =	6,	theta*=	0.188,	z =	46.863
[jh_simplexP]	Iteracio	15	:	iout =	7,	q =	12,	B(p) =	3,	theta*=	0.864,	z =	32.430
[jh_simplexP]	Iteracio	16	:	iout =	4,	q =	13,	B(p) =	24,	theta*=	0.640,	z =	25.819
[jh_simplexP]	Iteracio	17	:	iout =	-	q =	-	B(p) =	-	theta*=	-	z =	0.000

[jh_simplexP] Solucio basica factible trobada, iteracio 17

[jh_simplexP] Fase II

[jh_simplexP]	Iteracio	18	:	iout =	4,	q =	14,	B(p) =	13,	theta*=	0.065,	z =	183.243
[jh_simplexP]	Iteracio	19	:	iout =	2,	q =	6,	B(p) =	5,	theta*=	0.572,	z =	163.725
[jh_simplexP]	Iteracio	20	:	iout =	1,	q =	15,	B(p) =	7,	theta*=	374.780,	z =	145.879
[jh_simplexP]	Iteracio	21	:	iout =	9,	q =	16,	B(p) =	9,	theta*=	111.238,	z =	51.560
[jh_simplexP]	Iteracio	22	:	iout =	7,	q =	5,	B(p) =	12,	theta*=	0.618,	z =	7.871
[jh_simplexP]	Iteracio	23	:	iout =	5,	q =	17,	B(p) =	10,	theta*=	100.232,	z =	-5.342
[jh_simplexP]	Iteracio	24	:	iout =	10,	q =	7,	B(p) =	8,	theta*=	0.628,	z =	-30.623
[jh_simplexP]	Iteracio	25	:	iout =	7,	q =	12,	B(p) =	5,	theta*=	0.181,	z =	-46.593
[jh_simplexP]	Iteracio	26	:	iout =	10,	q =	9,	B(p) =	7,	theta*=	0.214,	z =	-57.585
[jh_simplexP]	Iteracio	27	:	iout =	1,	q =	19,	B(p) =	15,	theta*=	529.954,	z =	-60.506
[jh_simplexP]	Iteracio	28	:	iout =	3,	q =	8,	B(p) =	4,	theta*=	0.465,	z =	-172.346
[jh_simplexP]	Iteracio	29	:	iout =	3,	q =	13,	B(p) =	8,	theta*=	1.365,	z =	-193.197
[jh_simplexP]	Iteracio	30	:	iout =	-	q =	-	B(p) =	-	theta*=	-	z =	-197.476

[jh_simplexP] Solucio optima trobada, iteracio 30, z = -197.476081

[jh_simplexP] Fi simplex primal

Solucio optima:

```
vb = 19  6 13 14 17  2 12 11 16  9
xb =  651.2    3.4    1.4    0.4 433.8    6.6    0.8    2.9 1038.4    0.2
z = -197.4761
r =  152.8    0.0  43.4    25.9  52.8    0.0    2.4    9.2   -0.0 166.9    0.0    0.0    0.0
```

Problema 2

[jh_simplexP] Inici simplex primal amb regla de Bland

[jh_simplexP] Fase I

[jh_simplexP]	Iteracio	1	:	iout =	2,	q =	1,	B(p) =	22,	theta*=	0.377,	z =	2625.000
[jh_simplexP]	Iteracio	2	:	iout =	2,	q =	2,	B(p) =	1,	theta*=	0.684,	z =	2498.014
[jh_simplexP]	Iteracio	3	:	iout =	6,	q =	4,	B(p) =	26,	theta*=	0.166,	z =	2334.211
[jh_simplexP]	Iteracio	4	:	iout =	3,	q =	1,	B(p) =	23,	theta*=	0.175,	z =	2143.096
[jh_simplexP]	Iteracio	5	:	iout =	1,	q =	3,	B(p) =	21,	theta*=	0.384,	z =	2040.842
[jh_simplexP]	Iteracio	6	:	iout =	7,	q =	5,	B(p) =	27,	theta*=	0.864,	z =	1945.333
[jh_simplexP]	Iteracio	7	:	iout =	4,	q =	6,	B(p) =	24,	theta*=	1.436,	z =	1801.254
[jh_simplexP]	Iteracio	8	:	iout =	9,	q =	7,	B(p) =	29,	theta*=	1.060,	z =	1641.509
[jh_simplexP]	Iteracio	9	:	iout =	8,	q =	9,	B(p) =	28,	theta*=	0.162,	z =	855.444
[jh_simplexP]	Iteracio	10	:	iout =	5,	q =	8,	B(p) =	25,	theta*=	1.387,	z =	717.123
[jh_simplexP]	Iteracio	11	:	iout =	6,	q =	10,	B(p) =	4,	theta*=	0.473,	z =	184.750
[jh_simplexP]	Iteracio	12	:	iout =	5,	q =	11,	B(p) =	8,	theta*=	1.371,	z =	156.310
[jh_simplexP]	Iteracio	13	:	iout =	1,	q =	12,	B(p) =	3,	theta*=	0.069,	z =	49.013
[jh_simplexP]	Iteracio	14	:	iout =	4,	q =	4,	B(p) =	6,	theta*=	0.256,	z =	39.014
[jh_simplexP]	Iteracio	15	:	iout =	9,	q =	3,	B(p) =	7,	theta*=	0.230,	z =	18.269
[jh_simplexP]	Iteracio	16	:	iout =	10,	q =	8,	B(p) =	30,	theta*=	1.620,	z =	13.140
[jh_simplexP]	Iteracio	17	:	iout =	-	q =	-	B(p) =	-	theta*=	-	z =	0.000

[jh_simplexP] Solucio basica factible trobada, iteracio 17

[jh_simplexP] Fase II

[jh_simplexP]	Iteracio	18	:	iout =	6,	q =	6,	B(p) =	10,	theta*=	0.206,	z =	-246.297
[jh_simplexP]	Iteracio	19	:	iout =	9,	q =	7,	B(p) =	3,	theta*=	0.339,	z =	-519.979
[jh_simplexP]	Iteracio	20	:	iout =	6,	q =	14,	B(p) =	6,	theta*=	0.357,	z =	-537.351
[jh_simplexP]	Iteracio	21	:	iout =	7,	q =	3,	B(p) =	5,	theta*=	0.181,	z =	-557.947
[jh_simplexP]	Iteracio	22	:	iout =	6,	q =	15,	B(p) =	14,	theta*=	47.146,	z =	-562.608
[jh_simplexP]	Iteracio	23	:	iout =	6,	q =	20,	B(p) =	15,	theta*=	14.253,	z =	-580.288
[jh_simplexP]	Iteracio	24	:	iout =	-	q =	-	B(p) =	-	theta*=	-	z =	-586.610

[jh_simplexP] Solucio optima trobada, iteracio 24, z = -586.610343

[jh_simplexP] Fi simplex primal

Solucio optima:

vb = 12 2 1 4 11 20 3 9 7 8

xb = 3.1 0.7 1.8 1.5 1.2 14.3 0.3 1.5 0.0 3.4

z = -586.6103

r = 0.0 0.0 -0.0 -0.0 40.5 304.0 -0.0 0.0 -0.0 248.2 0.0 0.0 83.2

Problema 3

[jh_simplexP] Inici simplex primal amb regla de Bland

[jh_simplexP] Fase I

[jh_simplexP]	Iteracio	1	:	iout =	8,	q =	1,	B(p) =	28,	theta*=	0.586,	z =	1801.000
[jh_simplexP]	Iteracio	2	:	iout =	8,	q =	3,	B(p) =	1,	theta*=	0.667,	z =	1618.103
[jh_simplexP]	Iteracio	3	:	iout =	9,	q =	5,	B(p) =	29,	theta*=	0.363,	z =	1560.333
[jh_simplexP]	Iteracio	4	:	iout =	5,	q =	10,	B(p) =	25,	theta*=	0.556,	z =	1501.739
[jh_simplexP]	Iteracio	5	:	iout =	5,	q =	15,	B(p) =	10,	theta*=	58.814,	z =	1472.841
[jh_simplexP]	Iteracio	6	:	iout =	7,	q =	17,	B(p) =	27,	theta*=	82.953,	z =	1442.925
[jh_simplexP]	Iteracio	7	:	iout =	9,	q =	4,	B(p) =	5,	theta*=	0.660,	z =	1359.973
[jh_simplexP]	Iteracio	8	:	iout =	8,	q =	10,	B(p) =	3,	theta*=	0.422,	z =	1343.945
[jh_simplexP]	Iteracio	9	:	iout =	5,	q =	19,	B(p) =	15,	theta*=	19.867,	z =	1342.596
[jh_simplexP]	Iteracio	10	:	iout =	9,	q =	3,	B(p) =	4,	theta*=	0.080,	z =	1340.436
[jh_simplexP]	Iteracio	11	:	iout =	10,	q =	20,	B(p) =	30,	theta*=	56.709,	z =	1340.237
[jh_simplexP]	Iteracio	12	:	iout =	9,	q =	4,	B(p) =	3,	theta*=	0.085,	z =	1283.528
[jh_simplexP]	Iteracio	13	:	iout =	-,	q =	-,	B(p) =	-,	theta*=	-,	z =	1282.453
[jh_simplexP]	Problema infactible.												

Problema 4

[jh_simplexP] Inici simplex primal amb regla de Bland

[jh_simplexP] Fase I

[jh_simplexP]	Iteracio	1	:	iout =	8,	q =	1,	B(p) =	32,	theta* =	8.866,	z =	6828.000
[jh_simplexP]	Iteracio	2	:	iout =	5,	q =	2,	B(p) =	29,	theta* =	4.076,	z =	2483.670
[jh_simplexP]	Iteracio	3	:	iout =	5,	q =	3,	B(p) =	2,	theta* =	1.467,	z =	2397.856
[jh_simplexP]	Iteracio	4	:	iout =	2,	q =	4,	B(p) =	26,	theta* =	2.368,	z =	2050.292
[jh_simplexP]	Iteracio	5	:	iout =	4,	q =	5,	B(p) =	28,	theta* =	3.489,	z =	1347.373
[jh_simplexP]	Iteracio	6	:	iout =	2,	q =	2,	B(p) =	4,	theta* =	0.034,	z =	875.187
[jh_simplexP]	Iteracio	7	:	iout =	2,	q =	7,	B(p) =	2,	theta* =	0.102,	z =	813.890
[jh_simplexP]	Iteracio	8	:	iout =	2,	q =	8,	B(p) =	7,	theta* =	0.162,	z =	813.641
[jh_simplexP]	Iteracio	9	:	iout =	6,	q =	9,	B(p) =	30,	theta* =	0.954,	z =	783.651
[jh_simplexP]	Iteracio	10	:	iout =	10,	q =	4,	B(p) =	34,	theta* =	0.052,	z =	506.720
[jh_simplexP]	Iteracio	11	:	iout =	10,	q =	6,	B(p) =	4,	theta* =	0.078,	z =	489.532
[jh_simplexP]	Iteracio	12	:	iout =	10,	q =	7,	B(p) =	6,	theta* =	0.165,	z =	485.329
[jh_simplexP]	Iteracio	13	:	iout =	2,	q =	10,	B(p) =	8,	theta* =	0.251,	z =	483.995
[jh_simplexP]	Iteracio	14	:	iout =	2,	q =	11,	B(p) =	10,	theta* =	0.158,	z =	481.791
[jh_simplexP]	Iteracio	15	:	iout =	10,	q =	4,	B(p) =	7,	theta* =	1.330,	z =	451.599
[jh_simplexP]	Iteracio	16	:	iout =	1,	q =	6,	B(p) =	25,	theta* =	1.977,	z =	412.496
[jh_simplexP]	Iteracio	17	:	iout =	7,	q =	12,	B(p) =	31,	theta* =	0.711,	z =	248.833
[jh_simplexP]	Iteracio	18	:	iout =	5,	q =	10,	B(p) =	3,	theta* =	0.782,	z =	204.689
[jh_simplexP]	Iteracio	19	:	iout =	9,	q =	13,	B(p) =	33,	theta* =	0.592,	z =	187.369
[jh_simplexP]	Iteracio	20	:	iout =	3,	q =	2,	B(p) =	27,	theta* =	0.071,	z =	7.419
[jh_simplexP]	Iteracio	21	:	iout =	-	q =	-	B(p) =	-	theta* =	-	z =	0.000

[jh_simplexP] Solucio basica factible trobada, iteracio 21

[jh_simplexP] Fase II

[jh_simplexP]	Iteracio	22	:	iout =	3,	q =	7,	B(p) =	2,	theta* =	0.102,	z =	-788.970
[jh_simplexP]	Iteracio	23	:	iout =	8,	q =	8,	B(p) =	1,	theta* =	0.999,	z =	-791.546
[jh_simplexP]	Iteracio	24	:	iout =	7,	q =	14,	B(p) =	12,	theta* =	0.473,	z =	-876.151
[jh_simplexP]	Iteracio	25	:	iout =	10,	q =	2,	B(p) =	4,	theta* =	0.153,	z =	-908.193
[jh_simplexP]	Iteracio	26	:	iout =	10,	q =	17,	B(p) =	2,	theta* =	110.029,	z =	-915.046
[jh_simplexP]	Iteracio	27	:	iout =	4,	q =	12,	B(p) =	5,	theta* =	0.045,	z =	-934.298
[jh_simplexP]	Iteracio	28	:	iout =	2,	q =	18,	B(p) =	11,	theta* =	83.214,	z =	-941.734
[jh_simplexP]	Iteracio	29	:	iout =	7,	q =	4,	B(p) =	14,	theta* =	0.084,	z =	-975.353
[jh_simplexP]	Iteracio	30	:	iout =	3,	q =	15,	B(p) =	7,	theta* =	7.563,	z =	-979.763
[jh_simplexP]	Iteracio	31	:	iout =	7,	q =	19,	B(p) =	4,	theta* =	18.090,	z =	-981.786
[jh_simplexP]	Iteracio	32	:	iout =	4,	q =	11,	B(p) =	12,	theta* =	0.546,	z =	-993.564
[jh_simplexP]	Iteracio	33	:	iout =	4,	q =	16,	B(p) =	11,	theta* =	75.522,	z =	-1017.486
[jh_simplexP]	Iteracio	34	:	iout =	10,	q =	4,	B(p) =	17,	theta* =	0.182,	z =	-1025.916
[jh_simplexP]	Iteracio	35	:	iout =	9,	q =	20,	B(p) =	13,	theta* =	75.212,	z =	-1025.996
[jh_simplexP]	Iteracio	36	:	iout =	10,	q =	2,	B(p) =	4,	theta* =	1.770,	z =	-1099.114
[jh_simplexP]	Iteracio	37	:	iout =	8,	q =	17,	B(p) =	8,	theta* =	234.920,	z =	-1136.466

```

[jh_simplexP] Iteracio 38 : iout = 10, q = 21, B(p) = 2, theta*= 46.640, z = -1290.786
[jh_simplexP] Iteracio 39 : iout = 6, q = 11, B(p) = 9, theta*= 3.879, z = -1341.326
[jh_simplexP] Iteracio 40 : iout = 6, q = 22, B(p) = 11, theta*= 1170.705, z = -1410.966
[jh_simplexP] Iteracio 41 : iout = 5, q = 2, B(p) = 10, theta*= 16.187, z = -2293.986
[jh_simplexP] Iteracio 42 : iout = 9, q = 13, B(p) = 20, theta*= 3.701, z = -2384.056
[jh_simplexP] Iteracio 43 : iout = 5, q = 23, B(p) = 2, theta*= 167.687, z = -2725.466
[jh_simplexP] Iteracio 44 : iout = 7, q = 20, B(p) = 19, theta*= 725.863, z = -2962.166
[jh_simplexP] Iteracio 45 : iout = 1, q = 2, B(p) = 6, theta*= 1.983, z = -4267.536
[jh_simplexP] Iteracio 46 : iout = 1, q = 24, B(p) = 2, theta*= 132.077, z = -4486.566
[jh_simplexP] Iteracio 47 : iout = 9, q = 4, B(p) = 13, theta*= 799.000, z = -5777.366
[jh_simplexP] Iteracio 48 : problema no acotat detectat.
[jh_simplexP] Problema no acotat.

```